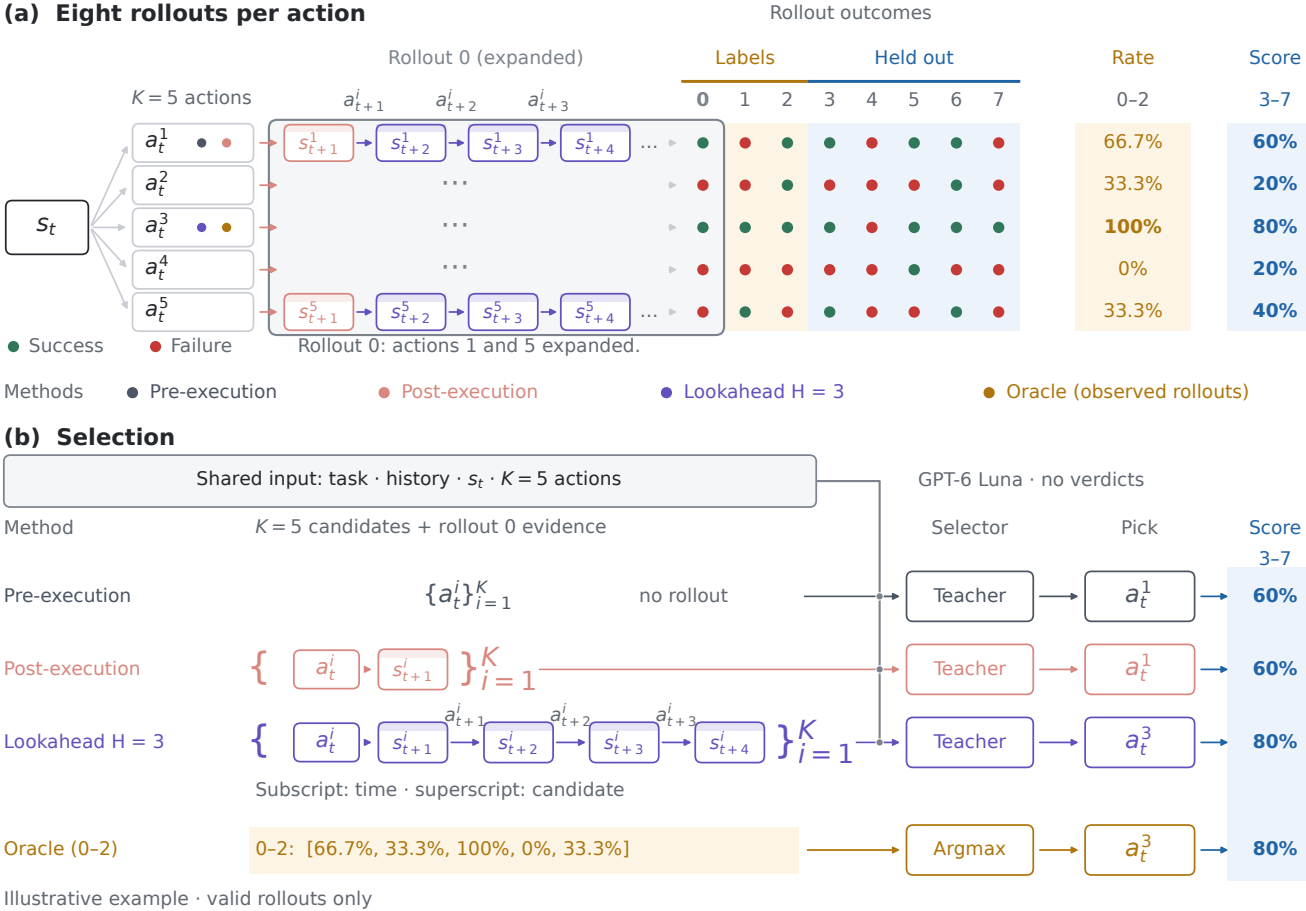
(a) Eight rollouts per action
Rollout outcomes
Rollout 0 (expanded)
Labels
Held out
Rate
Score
$K = 5$ actions
a^i_{t+1} a^i_{t+2} a^i_{t+3}
0 1 2 3 4 5 6 7
0–2
3–7
s_t
a^1_t a^2_t a^3_t a^4_t a^5_t
s^1_{t+1} s^1_{t+2} s^1_{t+3} s^1_{t+4} …
s^5_{t+1} s^5_{t+2} s^5_{t+3} s^5_{t+4} …
66.7% 60%
33.3% 20%
100% 80%
0% 20%
33.3% 40%
Success
Failure
Rollout 0: actions 1 and 5 expanded.
Methods
Pre-execution
Post-execution
Lookahead H = 3
Oracle (observed rollouts)
(b) Selection
Shared input: task · history · s_t · $K = 5$ actions
GPT-6 Luna · no verdicts
Method
$K = 5$ candidates + rollout 0 evidence
Selector
Pick
Score
3–7
Pre-execution
$\{a^i_t\}^K_{i=1}$
no rollout
Teacher
a^1_t
60%
Post-execution
$\{ a^i_t \to s^i_{t+1} \}^K_{i=1}$
Teacher
a^1_t
60%
Lookahead H = 3
$\{ a^i_t \to s^i_{t+1} \xrightarrow{a^i_{t+1}} s^i_{t+2} \xrightarrow{a^i_{t+2}} s^i_{t+3} \xrightarrow{a^i_{t+3}} s^i_{t+4} \}^K_{i=1}$
Teacher
a^3_t
80%
Subscript: time · superscript: candidate
Oracle (0–2)
0–2: [66.7%, 33.3%, 100%, 0%, 33.3%]
Argmax
a^3_t
80%
Illustrative example · valid rollouts only